HeartTrialTwin

Evidence-grounded heart-failure trial intelligence

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01 ABSTRACT

HeartTrialTwin integrates 2,824 heart-failure clinical trials from the AACT database into a source-linked knowledge graph to explore trial relationships, historical discontinuation, and prediction reliability. Community analysis identified 904 groups, with greater stability across repeated analyses but more isolated trials. Semantic methods outperformed GraphSAGE in association-ranking experiments. A separate evaluation of 329 trials yielded an ROC-AUC of 0.487 (95% CI: 0.385–0.589), failing to establish reliable predictive discrimination. These findings show how HeartTrialTwin can support evidence-grounded trial research while highlighting the challenges of predicting trial discontinuation.

02 RESEARCH PROBLEM

Predicting discontinuation in heart-failure trials is difficult because registry records change over time. These changes complicate trial comparisons, predictor selection, and evidence verification.

03 OBJECTIVE

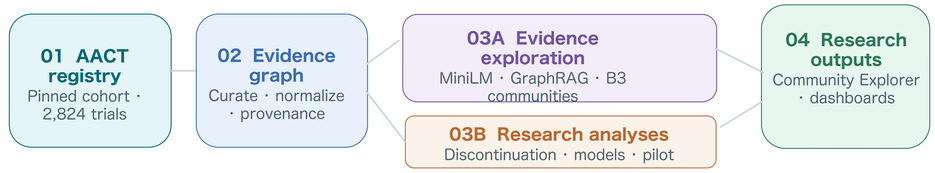
Develop HeartTrialTwin to link trial evidence, examine historical discontinuation, explore trial relationships, and assess predictive-model reliability.

04 METHOD AND COHORT SELECTION

A frozen cohort of 2,824 heart-failure trials from the September 5, 2026 AACT snapshot was used for retrospective analysis. Historical discontinuation patterns were examined across trial characteristics, while a separate evaluation assessed termination prediction using qualified archived records and a later status snapshot.
Outcome definitions: Historical discontinuation = Terminated or Withdrawn; V1 evaluation = Terminated versus Completed.
The full cohort was not used for prediction training. Completed status does not establish treatment effectiveness.

05 SYSTEM ARCHITECTURE / EVIDENCE WORKFLOW

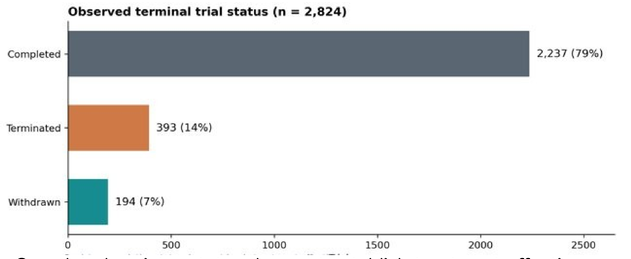
BUILD EVIDENCE → EXPLORE TRIALS → DESCRIBE HISTORY → EVALUATE PREDICTION



EVIDENCE CONTROLS · source provenance · citation checks · abstain if unsupported · research-only use

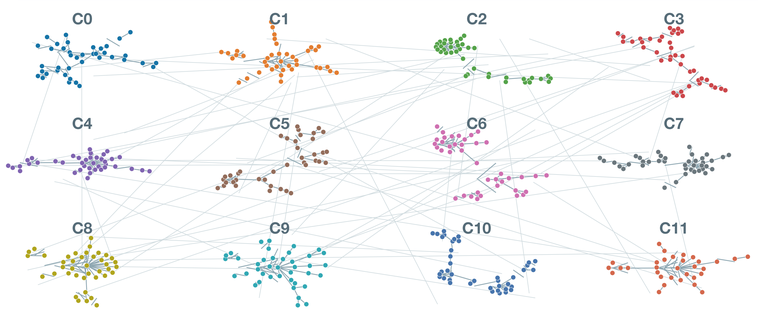
06 RESULTS

Recorded terminal status



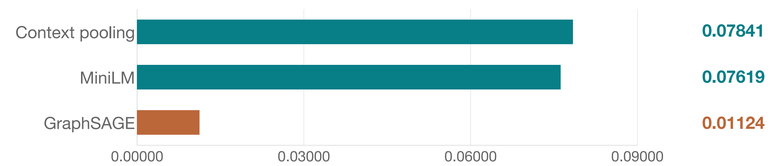
Completed registry status does not establish treatment effectiveness.

Exploratory trial communities (B3)



The network shows how 472 trials are grouped into 12 communities based on shared registered information. Only 521 of 10,270 relationships are displayed for clarity; these groups are exploratory, not clinical classifications.

Comparison of Trial-Association Ranking Methods



Context pooling and MiniLM performed better than GraphSAGE in ranking registered trial associations.
MRR (higher is better); 2,219 held-out queries and 2,824 candidate trials. This measures association ranking, not clinical prediction.

Community refinement

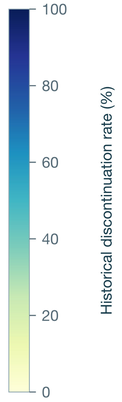
B1 and B3 are two versions of the trial-community construction approach, compared here for stability and fragmentation.

Saved comparison	B1	B3
Mean seed-to-seed ARI	**0.648**	**0.867**
Communities	**823**	**904**
Singletons	**747**	**820**

B3: higher stability, but more singleton groups.

Recorded discontinuation

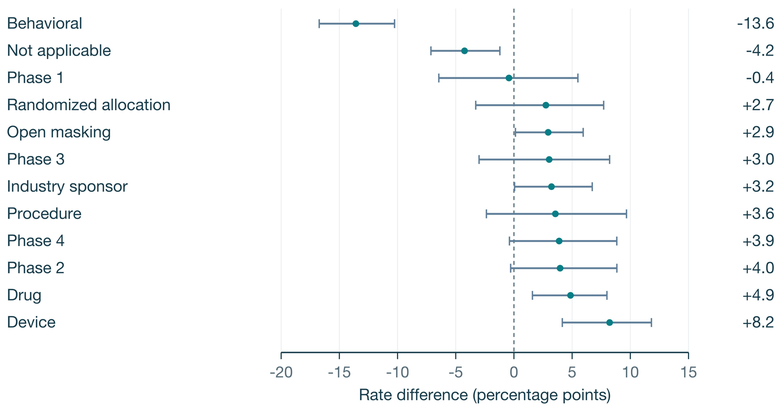
	Drug	Device	Behavioral	Procedure	Other	Biological
Early Phase 1	0/14	3/6	0/1	1/4	0/5	1/1
Phase 1	18%	3/19	—	5/11	4/14	5/11
Phase 1/2	21%	3/9	1/6	3/7	3/10	2/10
Phase 2	26%	28%	1/11	4/11	21%	5/19
Phase 2/3	12%	3/16	0/3	1/6	0/5	1/3
Phase 3	25%	26%	1/13	1/10	23%	0/4
Phase 4	25%	34%	0/11	4/13	18%	0/5
Not applicable	26%	27%	9%	21%	14%	1/5



Discontinued = Terminated + Withdrawn. Percentages for n ≥ 20; fractions for n < 20; — means no records. Intervention types overlap.

Discontinuation rates varied across trial phases and intervention types. These historical patterns do not predict future trial outcomes.

Historical factors and uncertainty



Unadjusted differences; exact family-cluster 95% CIs (500 saved resamples). Denominators vary. Enrollment contrast omitted because original-plan validity was unverified. Associations are not causal.

Separate V1 predictive evaluation

0.487

ROC-AUC, 95% CI 0.385–0.589

Saved status-snapshot assessment	Value
Binary-evaluable trials	**329**
Completed / Terminated	**289 / 40**
Average precision	**0.157**
Brier score	**0.1100**

358 scored; 29 Withdrawn excluded. Terminated versus Completed; event dates remain unverified. Not a validated prospective forecast.

The full research cohort contained 2,824 trials, but only 258 qualified trials were used for historical model development. A separate later-snapshot assessment included 329 evaluable trials.

Reliable predictive discrimination was not established.
Risk prediction release remains BLOCKED.

Evidence-grounded retrieval

Verified example: NCT00206232

Intervention: Spironolactone
Lead sponsor: Baylor College of Medicine
Seven registered planned outcomes, completely listed with citations.

Sources: interventions.txt, sponsors.txt, design_outcomes.txt. Saved validation traces preserve record IDs and citations.

Survival-benefit questions abstain: registered measures do not establish observed effects.

07 LIMITATIONS AND FUTURE WORK

Historical data: Registry records may have changed over time, limiting the reliability of information available at initial registration.
Prediction: The model was evaluated on a limited, qualified subset of trials and did not establish reliable predictive performance.
Future work: Verify historical records and trial chronology, expand the eligible cohort, and evaluate models on independent test cohorts.

08 CONCLUSION

HeartTrialTwin demonstrates the value of linking heart-failure clinical trials through a source-linked knowledge graph for evidence-grounded research. While the framework supports trial exploration and historical analysis, the evaluation did not establish reliable predictive performance.

09 ACKNOWLEDGMENT

I sincerely thank Prof. Yugyung Lee for her guidance and valuable feedback throughout the development of this project.

SCAN TO EXPLORE HEARTTRIALTWIN